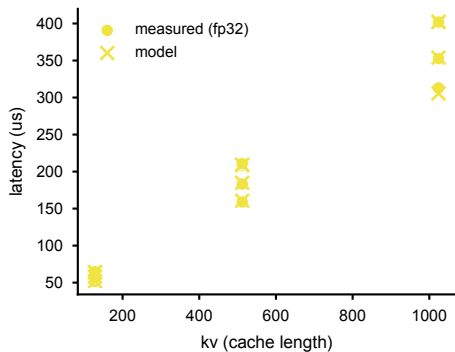
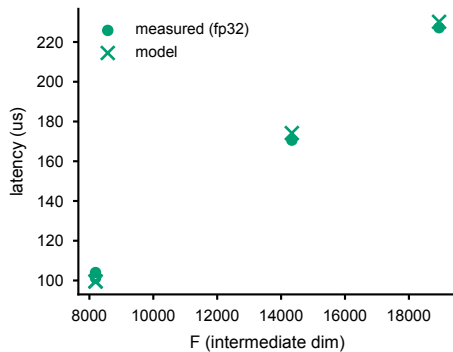


C1 — CPU instruction-count roofline: measured fp32 vs calibrated model (single A76 core; exp() = cost driver)

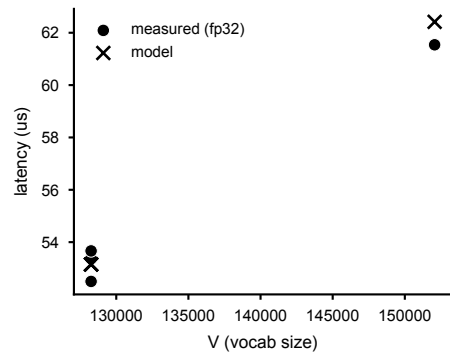
exp()-dominated: softmax



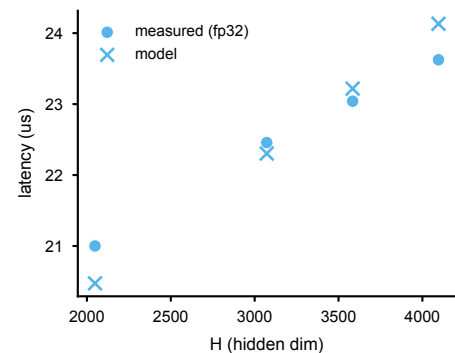
exp()-dominated: swiglu



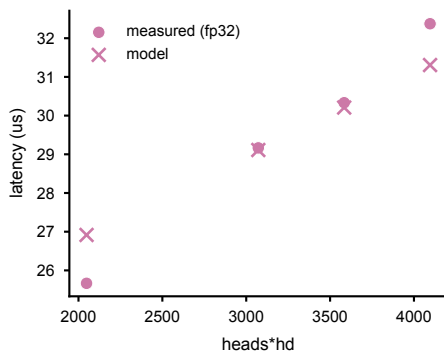
sampling_argmax



rmsnorm



rope_apply



residual

